Find and Fix: Inconclusive Tests and Forgotten Endpoints Answer Key

AP Calculus BC

Quick Answers

1. the limit of the terms = 0, —
2. the radius = 2, the value at the left endpoint = $-\ln(2)$, —
3. the corrected error bound = $\frac{1}{81}$, —
4. the corrected sum = $\frac{1}{12}$, —
5. the ratio-test limit = 1, the limit-comparison ratio = 1, —
6. the corrected radius = $\frac{1}{2}$, —
7. the corrected integral = 2, —
8. the value at the left endpoint = $-\ln(2)$, the term limit at the right endpoint = 0, —

What is verified

11 of 19 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC
LIM-7 – *problems 1, 3, 4, 5, 7*
LIM-8 – *problems 2, 6, 8*
Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

3. *If they got 0.0156: used the last term included instead of the first term omitted*
4. *If they got 1.333: used one as the first term instead of the term at n equal to two*
6. *If they got 1: read the radius off the shifted variable without factoring the two out*
7. *If they got 0.4: subtracted one from the exponent instead of adding one when antidifferentiating*

1. A student says $\lim 1/n = 0$, so $\sum 1/n$ converges by the nth-term test. (a) Confirm the limit. (b) Name the error.

Solution (a): The limit is correct: $\boxed{0}$

Solution (b) — instructor-judged. The nth-term test runs one way only. If the terms do *not* approach zero, the series diverges; if they do approach zero, the test proves nothing whatever. The student used it in the direction it does not have — and the series in front of them is the standard counterexample, since $\sum \frac{1}{n}$ diverges with terms going to zero. Full credit states the one-way rule and names the counterexample.

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2. $\sum x^n/(n2^n)$; a student reports the interval $(-2, 2)$. (a) Radius. (b) Sum at $x = -2$. (c) Correct interval.

Solution (a): $R = \lim \dfrac{2(n+1)}{n}$. $\boxed{R = 2}$

Solution (b): At $x = -2$ the terms become $\dfrac{(-2)^n}{n\,2^n} = \dfrac{(-1)^n}{n}$, the alternating harmonic series. $\boxed{-\ln 2 \approx -0.693}$

Solution (c) — instructor-judged. The ratio test settles only the open interval: at an endpoint its limit is exactly 1, so it is silent there by construction. Both endpoints must be substituted back. At $x = -2$ the series converges (alternating harmonic); at $x = 2$ it is $\sum \frac{1}{n}$ and diverges. The interval of convergence is $[-2, 2)$. Full credit names both endpoint series.

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3. $\sum (-1)^{n+1}/n^2$, first 8 terms; the student reports the bound $\frac{1}{64}$. (a) Correct bound. (b) Name the error.

Solution (a): The alternating series remainder is at most the first *omitted* term. Eight terms run to $n = 8$, so the first omitted term is $n = 9$, of size $\frac{1}{9^2}$. $\boxed{\frac{1}{81} \approx 0.0123}$

Solution (b) — instructor-judged. The student used the last term *included*, $a_8 = \frac{1}{64}$, instead of the first term *omitted*, $a_9 = \frac{1}{81}$. The bound reads $|S - S_N| \le a_{N+1}$ — the index is $N + 1$, not N. Full credit names the index and explains why the omitted term is the one that matters.

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4. A student sums $\sum_{n\ge 2} (1/4)^n$ as $\frac{4}{3}$. (a) Correct sum. (b) Name the error.

Solution (a): The first term actually written is the one at $n = 2$, namely $\left(\frac{1}{4}\right)^2 = \frac{1}{16}$, and the ratio is $\frac{1}{4}$:

$$S = \frac{1/16}{1 - 1/4} = \frac{1/16}{3/4}.$$

So the sum is $\boxed{\frac{1}{12} \approx 0.083}$

Solution (b) — instructor-judged. The geometric formula divides the *first term actually written* by $1 - r$, and the starting index decides what that first term is. The student used 1, which is the term at $n = 0$ — a term this sum does not contain. Full credit states the rule in a form that survives any starting index.

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5. A student says the ratio test gives 1 for $\sum \frac{n^2}{n^3+1}$, so it converges. (a) Confirm the ratio limit. (b) Limit comparison. (c) Name the error.

Solution (a): The ratio of consecutive terms is $\dfrac{(n+1)^2/((n+1)^3+1)}{n^2/(n^3+1)}$, whose limit is $\boxed{1}$

Solution (b): Comparing with $\frac{1}{n}$: $\dfrac{n^2/(n^3+1)}{1/n} = \dfrac{n^3}{n^3+1}$, whose limit is $\boxed{1}$

Solution (c) — instructor-judged. A ratio limit of 1 is the ratio test declining to answer — nothing at all had been established, so "so it converges" has no support. Worse, the failure was predictable: for terms that are a ratio of polynomials the ratio test always returns 1. Limit comparison with the harmonic series gives a finite positive limit, and since $\sum \frac{1}{n}$ diverges, so does this series. Full credit gives both the correction and the reason the ratio test had to fail.

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6. $\sum (2x-6)^n/n$; a student reports radius 1 and interval $(2,4)$. (a) Correct radius. (b) Both faults, and the correct interval.

Solution (a): With $(2x-6)^n = 2^n(x-3)^n$ the coefficients are $c_n = 2^n/n$, so $R = \lim \dfrac{n+1}{2n}$. $\boxed{R = \frac{1}{2}}$

Solution (b) — instructor-judged. Two faults. First, the coefficient 2 on the variable was never factored out, so the radius came out twice too large; the centre is 3 and the radius is $\frac{1}{2}$. Second, the endpoints were never tested. At $x = \frac{7}{2}$, $2x-6 = 1$ and the series is harmonic — diverges. At $x = \frac{5}{2}$, $2x-6 = -1$ and it is alternating harmonic — converges. The interval is $\left[\frac{5}{2}, \frac{7}{2}\right)$. Full credit names both faults.

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7. A student writes $\int_1^\infty x^{-3/2}dx = \frac{2}{5}$ and calls that the sum of $\sum n^{-3/2}$. (a) Correct integral. (b) Both errors.

Solution (a): Antidifferentiating adds one to the exponent: $\int x^{-3/2}dx = \dfrac{x^{-1/2}}{-1/2} = -\dfrac{2}{\sqrt{x}}$, so

$$\int_1^\infty x^{-3/2}\,dx = \left[-\frac{2}{\sqrt{x}}\right]_1^\infty = 0 + 2.$$

The integral equals $\boxed{2}$

Solution (b) — instructor-judged. Two errors. (i) The exponent was decreased rather than increased: $-\frac{3}{2} - 1 = -\frac{5}{2}$ appears in the student's antiderivative where $-\frac{3}{2} + 1 = -\frac{1}{2}$ belongs. (ii) The integral test decides convergence but its value is not the series' sum; the sum lies between the integral and the integral plus the first term, here between 2 and 3. Full credit names both.

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8. $\sum (x+2)^n/(n3^n)$; a student stops at $(-5,1)$. (a) Sum at $x=-5$. (b) Term limit at $x=1$. (c) Correct interval.

Solution (a): At $x=-5$, $x+2=-3$, so the terms become $\dfrac{(-3)^n}{n\,3^n} = \dfrac{(-1)^n}{n}$. $\boxed{-\ln 2 \approx -0.693}$

Solution (b): At $x=1$, $x+2=3$, so the terms become $\frac{1}{n}$, whose limit is $\boxed{0}$

Solution (c) — instructor-judged. The student stopped at the open interval the ratio test produced. Testing: $x=-5$ gives the alternating harmonic series, which converges, so that endpoint is included; $x=1$ gives the harmonic series, which diverges, so that endpoint is not. The interval is $[-5,1)$. Note the trap at the right endpoint: the terms there *do* approach zero, so the nth-term test says nothing and the harmonic series has to be recognised for what it is. Full credit names that trap.